Whole-Tree Message Passing with Local Clade-Change Features for Ancestry Inference

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Abstract

Changing inferred genealogies provide a structured representation for local ancestry prediction, but high span accuracy can conceal poor tract boundaries. We study a whole-tree graph neural pipeline on 476 diploids, with 32 training and eight development validation simulations. The model combines observed SNP features, complete PyTorch Geometric tree sweeps and a bidirectional genomic GRU. We compare local clade-change features, contiguous clade episodes and three ordered cut-memory rules. In the completed four-arm memory comparison, stable-leaf memory reaches $97.0778 \pm 0.5674\%$ present-day span accuracy, versus $97.0000 \pm 0.4978\%$ for continued training without added memory. Pair-cell memory reaches $97.0640 \pm 0.5595\%$ and fails both registered utility criteria against its controls. Stable-leaf boundary F1 is $18.3807 \pm 1.3141\%$, below continuation's $19.0488 \pm 1.4275\%$ at 1 kb tolerance. Current native ARGMix reaches $94.4461 \pm 1.1491\%$, with different input evidence, training and checkpoint rules. These reused-validation results distinguish span improvement from boundary quality and higher-order memory utility. They do not establish a new operator or overall benchmark superiority.

1 Introduction

Local ancestry inference assigns source-population labels along a sampled genome. A genealogy can organize evidence from many reference and query haplotypes, while its changes along the genome describe a second source of structure. These two axes pose different modeling problems. Within a marginal tree, information must travel between known references and queries through anonymous inferred ancestors. Across trees, sampled haplotypes persist while internal nodes and descendant clades change.

Recent ancestry systems motivate learning from genealogical structure. AncestralPaths accompanies a demographic analysis of ancient Eurasians (Irving-Pease et al. 2024). ARGMix uses graph-transformer representations and genomic sequence modeling (Shanks et al. 2026). ARGformer learns path representations and performs native retrieval (Bonet et al. 2026). Their released representations, supervision and prediction rules differ. A comparison on common observations therefore needs both a common scoring endpoint and an explicit account of the remaining pipeline differences.

We investigate a compact whole-tree model that retains all 952 observed haplotypes and every inferred marginal tree. Its graph computation uses PyTorch Geometric (Fey and Lenssen 2019). A bidirectional GRU (Cho et al. 2014) models the complete sequence of tree contexts for each sampled haplotype. Observed SNP features and reference annotations enter the graph encoder directly. Predictions do not consume another ancestry model's outputs.

The empirical question is whether coupling learned states across changing trees is useful beyond simpler structural features. We first compare tensor clade-event interactions with matched event-MLP and node-local flag controls. A later completed H045 comparison pools states across consecutive occurrences of the same proper clade and compares this foreign context with an equal-parameter own-state arm. The latter computes the same episode pool but uses its own state in the residual. This matched comparison separates a foreign-state effect from the capacity and additional training shared by the two arms.

The completed comparisons distinguish three effects. H045 improves the development mean over earlier conditions, but its foreign-episode arm exceeds own-state by only 0.0096 percentage points. A further H050 four-arm continuation reaches higher span scores, with stable-leaf memory ahead of pair-cell memory and the continued-training control strongest on boundary F1. Pair-cell memory fails its registered utility criteria against all three controls. High span accuracy therefore neither establishes a higher-order memory advantage nor resolves tract-boundary prediction. A fixed-output diagnostic separates missing candidate boundaries from errors at boundaries the decoder could express.

The study uses independent simulations as the unit of partitioning and reporting. Its eight validation simulations also support checkpoint selection and architecture development. The reported variation is across those simulations from one optimizer seed per condition. Independent confirmation, broader comparators and a completed novelty assessment remain necessary before stronger claims.

2 Task and Evaluation Protocol

2.1 Observed Panel and Simulation Split

The common panel contains 476 diploids, or 952 phased haplotypes. There are 686 query haplotypes and 266 pure-

reference haplotypes. Queries comprise present-day individuals (182 haplotypes), Bronze Age (162), BAA (14), Neo (302) and Yam (26). The reference panel comprises WHG (102), EHG (86), Ana (50) and CHG (28). Cohort names follow the released roster. In particular, Bronze Age is only an approximate correspondence to the ARGMix paper's Post-Neolithic category.

Each independent simulation spans 2 Mb. The completed development data contain 32 TRAIN and eight VAL simulations. Simulation identity is assigned before downstream windows, sampled rows or genomic chunks are constructed. Thus nearby trees from a single simulation do not enter different partitions. An independent outer assessment is outside the results reported here.

The simulation condition follows the released ancient-European demographic and sample-panel definition, with a uniform recombination rate of 10^{-8} and mutation rate of 1.25×10^{-8} per base per generation. Current msprime (Baumdicker et al. 2022) settings use the Hudson ancestry model, discrete coordinates and JC69 mutations. Genealogies used by the present full-panel models are native Relate post-population-size reconstructions (Speidel et al. 2019). Reused source-module names containing *tsinfer* do not identify the topology actually read in this experiment.

Observed inputs include phased genotypes, inferred topology and branch ages, sample ages and roles, and known reference-source classes. Query ancestry annotations are external supervision. Census labels, truth breakpoints and query-source labels do not define the model's observed feature bank, genomic holds or evaluation coordinates. Reference labels are available to every applicable native adapter as observed panel annotations.

2.2 Common Four-Class Endpoint

The shared endpoint has class order Ana, CHG, WHG and EHG, denoted C4. The native AncestralPaths classifier produces six path probabilities. For the common endpoint we collapse them before choosing a class,

$$(p_{\mathrm{Ana}}, p_{\mathrm{CHG}}, p_{\mathrm{WHG}}, p_{\mathrm{EHG}}) = (p_1 + p_5, p_2 + p_6, p_3, p_4). \tag{1}$$

Its native six-path SNP-focal endpoint remains separate. This mapping defines the present comparison and does not reconstruct a separately trained model from a published table.

Scoring uses the same original 100 bp center-bin support for every pipeline. Bins are clipped to inferred coverage. Missing support is not filled and uncovered tails are not extrapolated. Truth intervals are intersected with these fixed bins to obtain exact class-specific physical mass. Let m_{iqc} be the represented base-pair mass of class c for bin i and query q, and let p_{iqc} be a model probability. For a cohort Q,

$$A_Q = \frac{\sum_{i,q\in Q,c} m_{iqc} \mathbf{1}\{c = \arg\max_k p_{iqk}\}}{\sum_{i,q\in Q,c} m_{iqc}}, \tag{2}$$

$$\mathrm{NLL}_Q = -\frac{\sum_{i,q\in Q,c} m_{iqc} \log p_{iqc}}{\sum_{i,q\in Q,c} m_{iqc}}. \tag{3}$$

Zero probability on positive true-class mass yields infinite NLL. We apply no probability floor, post-hoc renormalization, purity filter or fitted calibration.

The primary endpoint is equal-simulation mean accuracy for the 182 present-day query haplotypes. All-query, ancient-query and individual-cohort results are secondary. We report sample SD with divisor $8 - 1$ across the eight independent VAL simulations. These SDs describe simulation variation for the fitted models, rather than optimizer-seed uncertainty or confidence intervals. Primary selection scores retain their registered FP32 arithmetic. Broader diagnostics integrate physical spans in FP64 and agree at the displayed precision.

3 Whole-Tree Model with Local Clade Flags

Let T_t denote the inferred tree on genomic hold I_t, with the same ordered set of sampled haplotypes in every tree. Internal-node indexes are local to a tree. The model makes a C4 probability vector for every query on each native hold. Figure 1 shows the implemented data flow.

3.1 Observed SNP and Node Encoding

For each SNP and sampled haplotype, the site encoder receives 19 observed channels: eight logged donor allele-frequency values for its called allele, eight availability masks, the log panel frequency of that allele, site entropy and a call indicator. Missing calls are masked. A $19 \to 32 \to 32$ MLP with GELU and tanh transforms each tuple. Within each native hold, the encoder pools the mean and coordinatewise maximum, then appends log call count, an empty-pool indicator and log hold span. A $67 \to 32$ projection produces a sampled-leaf SNP representation.

Known donor labels enter as a four-class one-hot vector. Query labels are masked and accompanied by an unknown-label indicator. The leaf representation also receives normalized log sample age. A $38 \to 32$ projection combines these six annotations with the SNP encoding. Every graph node receives a separate projection of normalized log node age, root status and sampled-node status. The sampled-leaf feature is added only at its own native node.

Two sigmoid gates, one for each message direction, are computed from nine observed edge channels. These combine inferred edge geometry with mutation-concordance summaries. All graph states have width 32. This encoding keeps known reference annotations distinct from the query targets used by the external loss.

3.2 Upward Messages and Recipient Exclusion

Each tree sweep uses actual PyG `MessagePassing` and `SumAggregation` calls. Let b_v be a node's encoded base feature, $g_u^{\uparrow}$ its upward gate and d_v the number of children of v. The upward messages and state have the form

$$m_{u\to v} = \frac{g_u^{\uparrow} \odot W_{\uparrow} h_u}{\max(1, d_v)}, \tag{4}$$

$$s_v = \sum_{u\in \mathrm{child}(v)} m_{u\to v}, \tag{5}$$

$$h_v = \mathrm{LN}(b_v + \tanh(A_{\uparrow}[b_v, s_v] + a_{\uparrow})). \tag{6}$$

Native topological levels determine evaluation order. The exact transformed upward contribution of every child is retained for the downward sweep.

For a nonroot node u with parent v, the downward context excludes that child's own upward contribution,

$$c_{v\setminus u} = o_v + b_v + s_v - m_{u\to v}. \tag{7}$$

A gated linear transformation, learned downward update, tanh and layer normalization map this context to o_u. Roots start with zero outside state. The sampled-leaf context is a learned width-32 projection of the concatenated upward and outside states. Both sweeps traverse every native node. They neither select a fixed ancestor depth nor replace the tree with a query-reference subgraph.

3.3 Exact Local Clade-Change Flags

For an internal node v, let $C_t(v)$ be its set of descendant sampled haplotypes. A proper clade contains more than one and fewer than 952 sampled leaves. The structural preparation compares these sets exactly between adjacent trees using temporary integer bitsets. Equal proper clades on unary chains use the deepest structural representative. Leaf and root clades are excluded.

A representative receives a gained flag g_v if its proper clade is absent from the preceding tree and a lost flag ℓ_v if absent from the following tree. The corresponding missing-neighbor flags at sequence ends remain zero. Only two compact flags per native node are retained. Exact bitsets are not neural features and no ancestral-node correspondence is asserted. In particular, an identical sampled-leaf clade is a structural statement, rather than proof that two inferred internal nodes are the same biological ancestor.

The H042 adapter operates on a node's own first upward state $h_v^{(1)}$,

$$\Delta_v = O\Big[g_v(Uh_v^{(1)} \odot Vh_v^{(1)}) - \ell_v(Uh_v^{(1)} \odot Wh_v^{(1)})\Big]. \tag{8}$$

Here $U, V, W : \mathbb{R}^{32} \to \mathbb{R}^8$ and $O : \mathbb{R}^8 \to \mathbb{R}^{32}$ are bias-free. Their 1,024 weights increase the body from 29,764 to 30,788 parameters. The output projection O starts at zero. A second complete tree sweep uses $b_v + \Delta_v$ with the original gates and the same sweep weights. Fresh state buffers are used for this second traversal.

Equation 8 contains no other node's or tree's latent state. It is a local structural-feature control for the more elaborate clade-event interactions, not a claimed novel message-passing primitive.

3.4 Matched Clade-Episode Context

H045 retains the complete learned H042 model and adds a paired episode adapter. A maximal consecutive run of the same exact proper sampled-leaf clade is an episode. Reappearance after an absent tree starts a new episode. The same unary representative rule applies, without asserting biological ancestor identity. For a member i with first-sweep state h_i and physical span w_i, the foreign context excludes its own tree,

$$m_i = \frac{\sum_{j\in E(i)} w_j h_j - w_i h_i}{\sum_{j\in E(i)} w_j - w_i}. \tag{9}$$

Only episodes containing at least two trees receive an extra residual,

$$\Delta_i^{\mathrm{ep}} = O_e[(U_e h_i) \odot (V_e m_i) + W_e h_i]. \tag{10}$$

The own-state arm substitutes $m_i = h_i$ on the same eligible nodes while computing the same pool. Three bias-free $32 \to 8$ projections and one $8 \to 32$ projection add 1,024 weights, for 31,812 per arm. The output projection starts at zero. The second shared tree sweep receives $b + \Delta + \Delta^{\mathrm{ep}}$. PyG sums implement the span-weighted pool, and gradients remain connected across episode members. This is an implemented conditioning comparison, not a claim of a new pooling or message-passing operator. Provenance: E13–E14.

3.5 Ordered Cut-Memory Comparison

H050 starts from H045 own-state and adds memory at observed tree cuts 350 generations before the present. Samples older than the cut retain their actual sampled-node singleton. At each cut representative, a bias-free $32 \to 8$ projection receives the sum of first-sweep upward and outside states. Forward and reverse memory scans use physical-span decay $\rho_t = \exp(-w_t/\ell)$ with a learned positive length scale initialized to 5 kb. The pair-cell arm transports histories through sample-witness intersections of successive cut groups. Its adjacent-marginal null retains only the corresponding middle-group marginal. A stable-leaf control instead carries memory along persistent sampled haplotypes. Each arm projects the bidirectional history minus the current forcing through an $8 \to 32$ map and injects the result at cut representatives before the second tree sweep. The output map starts at zero. The three arms add 513 parameters, for 32,325 each, and retain the original genomic GRU and direct classifier. A fourth arm continues the 31,812-parameter H045 model without adding memory. This comparison tests these specific memory rules using established aggregation and recurrence. Provenance: E16–E17.

3.6 Complete Genomic History and Readout

The second-sweep leaf contexts are concatenated with three observed geometry channels: log hold span and log distances to the preceding and following hold midpoints. The normalization denominator is $\log(2{,}000{,}001)$. A single-layer bidirectional GRU receives the resulting 35-channel sequence, with hidden width 32 in each direction. It processes all native holds for all 952 sampled haplotypes.

The classifier concatenates the local 32-channel context and 64-channel recurrent history. A $96 \to 32 \to 4$ MLP with GELU yields C4 logits. Only afterward does the readout select the 686 queries and map each original scored bin to its native hold. Consequently the current prediction is constant within a hold. SNP chunks of 256 are a memory strategy for exact pooling, not truncated recurrent histories or independent genomic training examples.

This construction combines established graph propagation, local structural features and recurrent sequence modeling. Temporal graph state-space methods provide a related perspective on persistent sampled interfaces (Li et al. 2024). The present empirical comparisons evaluate these

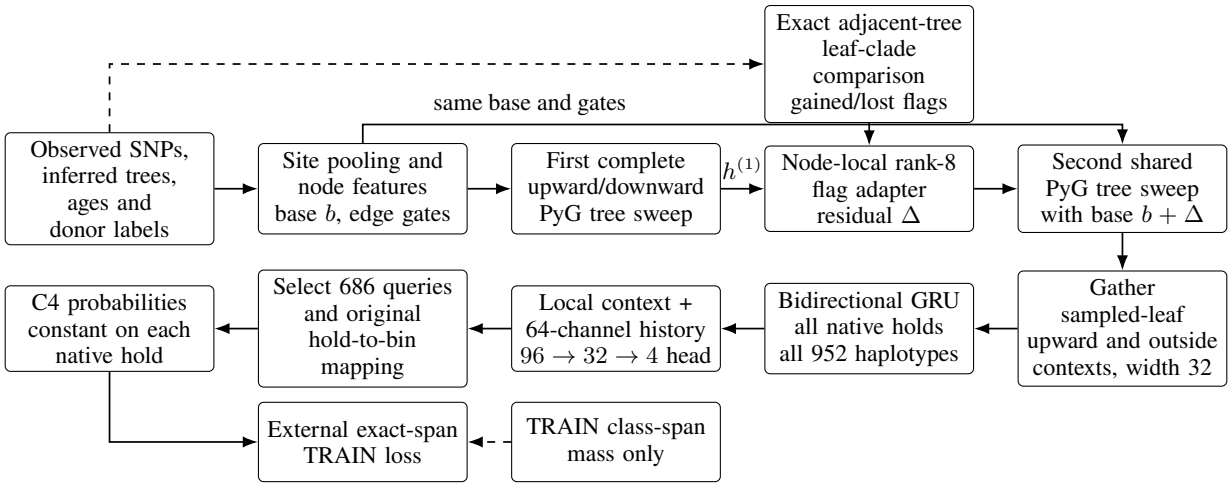


Figure 1: The inherited H042 computation. The two full-tree sweeps share learned weights. Only first upward states feed the local adapter. Dashed arrows supply structural flags or external training supervision. The H045 comparison adds the episode residual in Section 3.4 before the same second sweep, while retaining the full genomic GRU. Query truth never enters the forward feature path. Architecture provenance is E6–E7 and E14 in Appendix E.

concrete implementations and do not resolve priority across the broader literature.

4 Training and Comparator Conditions

4.1 Whole-Parent Optimization

The training objective uses all 686 query haplotypes. Because logits are constant on a native hold, class-specific target spans from every original scored bin can be summed by hold without changing the physical-span cross-entropy objective. For hold t,

$$M_{tqc} = \sum_{i:\, r(i)=t} m_{iqc}, \qquad \mathcal{L} = -\frac{\sum_{tqc} M_{tqc} \log p_{tqc}}{\sum_{tqc} M_{tqc}}. \tag{11}$$

This aggregation preserves every represented target span. It does not sample trees or discard query states. Targets remain external to the observed bank.

The H038 aligned body and its zero-concordance control each complete 30 FP32 passes, or 960 whole-simulation optimizer updates. The zero-concordance arm sets the first three branch-support channels to zero. It retains SNP inputs, topology, remaining support summaries and the complete GRU. Thus it is a particular routing-feature ablation, not an SNP-free or genealogy-free comparator.

H040 tensor and event-MLP conditions and H042 flags begin from the same selected H038 aligned checkpoint. Each completes 30 additional passes and 960 additional updates. AdamW is reset with learning rate 10^{-3}, weight decay 10^{-4}, betas $(0.9, 0.999)$ and epsilon 10^{-8}. Gradient norm is clipped at 1. These extensions have 30,788 parameters and use optimizer seed 101. The same added budget does not make their total exposure equal to the unextended H038 body.

H045 begins from the same selected H042 pass-30 checkpoint in both arms, preserves its complete flag adapter and trains all body weights. Its new episode weights are identical at initialization, and each arm independently resets the same AdamW settings and completes 30 FP32 passes, or 960 further whole-simulation updates, with seed 101. Each arm therefore has 1,920 continuation updates beyond the selected H038 base. This extra exposure and its 31,812 parameters must be distinguished from H042's 30,788 parameters. Provenance: E13–E14.

H050 initializes every arm from the completed H045 own-state pass-30 model, whose lineage contains 2,880 prior updates. Each arm resets AdamW, trains all weights and completes another 30 FP32 passes, or 960 updates, with seed 101. The four-arm experiment therefore adds 3,840 optimizer updates in total, and each completed model lineage reaches 3,840 updates. Memory arms select added pass 30, while continuation selects added pass 20, corresponding to 3,520 updates in that selected checkpoint's lineage. All arms finish the same training budget despite their different selected passes. All 192 checkpoint-parent prediction products are sealed before any VAL truth is opened. Provenance: E16.

Checkpoints at five-pass intervals are evaluated using equal-simulation present-day VAL accuracy, with lower NLL and earlier-pass tie breaks. The aligned H038 checkpoint is pass 30, while zero concordance selects pass 25 although both finish pass 30. H040 and H042 select additional pass 30. Both H045 arms also select their additional pass 30. All 96 H045 checkpoint-parent products are sealed before validation truth is opened for scoring. VAL supplies no optimization gradients, but repeated development and selection on this split limit confirmatory interpretation.

4.2 Matched Interactions and Native Baselines

The H040 comparison holds its clade-event inputs, initialization protocol, 1,024 added weights and additional update budget fixed between a tensor interaction and an event MLP. Both route signed residuals through changed-clade event in-

cidences, followed by the second shared tree sweep. The local H042 control uses the same size and added budget but substitutes the two exact local flags in Equation 8. It therefore asks whether those event interactions are needed beyond local topology-change information.

Native AncestralPaths retains the released Keras Dense512 classifier, six path outputs and fixed five-epoch training. Its actual fit uses 5,190,874 sampled rows from the 32 TRAIN simulations and selects final epoch 5. This is the executable released Dense model, not a reconstruction of a published CNN. Its derived C4 probabilities are scored without changing the native model.

Current native ARGMix retains its author implementation, with disclosed full-panel slot and metadata adaptation. It uses the released 15-reference representation without ancestor features and the native genomic sequence model. Its fit uses source seed 50 and selects epoch 9 by the original smoothed, masked, unweighted VAL batch-mean loss. Native patience is 2 with a 20-epoch cap. External full-support scoring does not replace that inner selection rule.

These conditions share simulation observations and physical scoring support, but they do not isolate architecture. The whole-tree models additionally consume observed SNP features. Native row sampling, reference representations, sequence chunks, supervision, training exposure and checkpoint objectives remain method-specific. H040 and H042 receive 960 additional updates after H038. H045 receives another 960 after H042, and H050 another 960 per arm after H045 own-state. Comparisons with native pipelines are therefore development pipeline comparisons, rather than matched-input or matched-budget operator tests.

5 Results

5.1 Primary Span Accuracy

Table 1 gives the completed present-day development results. H045 foreign-episode context reaches $96.8270 \pm 0.5870\%$, while its matched own-state control reaches $96.8175 \pm 0.6142\%$. Current native ARGMix reaches $94.4461 \pm 1.1491\%$ and the AP-derived C4 endpoint reaches $93.7830 \pm 0.8385\%$. The H045–ARGMix difference is 2.3809 percentage points between the specified fitted pipelines. Extra SNP inputs, continued training and different native checkpoint rules prevent attributing that gap to cross-tree exchange. The relevant matched H045 effect is only 0.0096 points.

The aligned H038 body exceeds zero concordance by 0.5639 percentage points at the selected checkpoints. Both runs complete 960 updates, but their selected passes differ. This result supports retaining the observed concordance features in this development condition. It does not identify an effect of all SNP information, all genealogy information, or PyG software alone.

The same caution applies to improvements over the H038 body. H042 changes capacity, performs two shared tree sweeps and trains for another 960 updates. H045 increases capacity again and continues the selected H042 model. These score differences cannot be assigned entirely to a structural mechanism. The closer controls match both added capacity and training within each extension family.

Condition	Accuracy (%)	Mean NLL
Native AP (C4)	93.7830 ± 0.8385	∞
Current native ARGMix	94.4461 ± 1.1491	0.219695
H038 zero concordance	94.9252 ± 0.7917	0.147232
H038 aligned	95.4891 ± 0.8338	0.133497
H040 event MLP	96.2007 ± 0.7048	0.110014
H040 tensor	96.2674 ± 0.7493	0.109332
H042 local flags	96.5778 ± 0.5954	0.100656
H045 own-state	96.8175 ± 0.6142	0.092976
H045 foreign episode	96.8270 ± 0.5870	0.092980

Table 1: Present-day development span accuracy, mean±sample SD across eight simulations and one optimizer seed per condition. NLL has no probability floor. Native AP is the derived C4 endpoint. H040/H042 add 960 updates after H038, and H045 adds another 960 after H042. Provenance: E1–E5, E11 and E13, Appendix E.

Paired comparison	Difference (pp)	Positive
Tensor − event MLP	0.0667 ± 0.0886	6/8
Flags − tensor	0.3104 ± 0.2588	8/8
Flags − event MLP	0.3771 ± 0.2165	8/8
H045 foreign − own	0.0096 ± 0.1527	6/8

Table 2: Equal-simulation paired accuracy differences, mean±sample SD. Positive counts are descriptive. Each comparison matches added parameter count and training within its family. H045 continues H042 and is a separate paired comparison. Provenance: E2–E3 and E13.

Native AncestralPaths has finite accuracy but infinite strict raw FP32 NLL on the primary cohort. One simulation places zero probability on the true class over 800 query-base pairs. We preserve this result rather than floor probabilities, exclude the simulation or silently average only finite NLLs. Accuracy and NLL measure different aspects of a prediction distribution.

5.2 Local Flags Versus Clade-Event Interactions

Table 2 reports paired present-day differences across the same eight simulations. Tensor coupling exceeds its event MLP by 0.0667 percentage points. H042 local flags exceed tensor coupling by 0.3104 points and the event MLP by 0.3771 points. In the later H045 family, foreign episode context exceeds own-state by 0.0096 ± 0.1527 points, positive in six of eight simulations. This fails the registered practical accuracy-gain criterion of at least 0.10 points. The gate is a development utility criterion, not a significance test.

The earlier comparisons favor local flags over the examined clade-event interactions. The later episode comparison does not establish a useful foreign-state advantage over its own-state control. Different incidence definitions, objectives or independent simulations could change these results. Neither the tensor interaction nor the episode pool is supported here as a distinct new neural operator.

H050 arm	Accuracy (%)	Boundary F1 (%)
Pair-cell memory	97.0640 ± 0.5595	18.93 ± 1.42
Marginal null	97.0504 ± 0.5796	18.85 ± 1.05
Stable leaves	97.0778 ± 0.5674	18.38 ± 1.31
Continuation	97.0000 ± 0.4978	19.05 ± 1.43

Table 3: H050 present-day results, mean±sample SD across eight reused VAL simulations and optimizer seed 101. Boundary tolerance is 1 kb. All arms complete 960 new updates from the same H045 own-state model. The memory arms have 32,325 parameters and select pass 30. Continuation has 31,812 parameters and selects pass 20. Provenance: E16.

5.3 Matched Ordered-Memory Ablation

Table 3 reports the completed H050 comparison on the same eight development simulations. Stable-leaf memory has the highest present-day accuracy of the four arms, but continuation has the highest boundary F1. Pair-cell memory exceeds the adjacent-marginal null by 0.0137 ± 0.0243 accuracy points and falls below stable leaves by 0.0138 ± 0.0384 points. Its gain over continuation is 0.0640 ± 0.1453 points. These are means and sample SDs of paired differences, distinct from the per-arm SDs in the table.

The registered primary criterion requires pair-cell memory to gain at least 0.10 accuracy points without reducing boundary F1 against each of the three controls. The secondary criterion requires at least 1.00 boundary-F1 point without losing more than 0.05 accuracy points against each control. Both criteria fail. Pair-cell memory's mean boundary F1 is 0.1224 points below continuation. Stable leaves gain 0.0778 accuracy points over continuation but lose 0.6682 boundary-F1 points. Thus this completed ablation does not support a useful higher-order memory advantage, despite the larger span scores than the earlier H045 checkpoints. These remain one-seed development results.

5.4 Cohort Dependence

Table 4 retains the earlier native and H042 cohort comparison, including the small BAA and Yam groups. Appendix Table 7 adds both completed H045 arms. Foreign-episode context reaches 98.7761% across all 686 queries, compared with 98.7815% for own-state. Its present-day score is 96.8270%. These aggregate and primary endpoints are not interchangeable, and the foreign condition does not uniformly exceed its matched control.

The contemporary-versus-ancient contrast reflects different query distributions under this demographic condition. Query cohorts differ in ages, source mixtures, tract structure and numbers of haplotypes. These results do not measure performance on an empirical ancient-DNA collection or establish robustness to observation error. Small cohorts also have sparse transition counts, so a high span score does not guarantee a stable boundary estimate.

Advantages are not uniform within the extended neural family. H040 tensor has BAA accuracy 99.14%, compared with H042's 99.02%. The event MLP has Neo accuracy 99.56%, compared with H042's 99.53%. Appendix A gives their simulation SDs. The primary ranking thus summarizes a particular registered endpoint rather than dominance on every cohort and metric.

5.5 High Span Accuracy, Weak Boundaries

Table 5 uses the unchanged one-to-one positional boundary matcher at 1 kb tolerance. H042 improves mean F1 to $18.05 \pm 1.84\%$, but its recall is only $22.30 \pm 1.86\%$ and precision is $15.21 \pm 2.01\%$. Accurate long tracts can dominate physical-span accuracy while numerous missed and extra transitions remain.

H045 foreign-episode boundary F1 is $18.1558 \pm 1.3790\%$, compared with $18.0557 \pm 1.7761\%$ for own-state. The paired difference is 0.1001 ± 0.9632 percentage points. It does not resolve the boundary weakness, and the failed accuracy-gain criterion means that H045 does not meet both registered utility criteria. These H045 results do not replace the separate H042 event counts or the H038 fixed-output diagnostic. Provenance: E13.

Native AP has higher boundary recall than H042 in this comparison, accompanied by much lower precision. H042's larger F1 consequently should not be read as an improvement in every transition metric. For H042 the pooled counts are 4,008 true changes, 5,920 predicted changes and 892 positional matches. These pooled event counts are distinct from the equal-simulation means and SDs in the table.

A separate fixed-output H038 diagnostic clarifies two limitations. Of 4,008 true changes, 1,094, or 27.30%, have no allowed native-hold readout edge within 1 kb. Its class-agnostic hold-edge oracle recall is $72.70 \pm 2.80\%$. Yet 2,195 of the aligned model's 3,289 missed boundaries still have a nearby candidate. Missing readout resolution therefore explains only part of the recognition deficit.

False internal ancestry islands of at most 10 kb account for $35.46 \pm 2.71\%$ of H038's extra boundaries, while all predicted tracts of that length contain only $3.83 \pm 0.73\%$ of wrong query-base pairs. Event counts and wrong physical mass emphasize different errors. This diagnostic concerns the frozen H038 outputs and is not an H042 island analysis. It motivates prospective continuity and finer-readout comparisons without selecting a smoothing rule on these VAL results.

5.6 Frozen Application to Ancient Danish Genomes

We applied the selected H045 foreign-episode checkpoint to a fixed panel of 18 ancient Danish individuals from published imputed genomes (Irving-Pease et al. 2024). Eight proxy individuals provide 16 labelled reference haplotypes. Ten query people comprise five earlier individuals, dated 5.44–5.67 ka cal BP, and five later individuals, dated 3.97–4.26 ka cal BP. Three fixed windows on chromosomes 20–22, each at $[20, 30)$ Mb, yield 1,158 native Relate trees and 8,587 qualified site records. Features use original phased genotype calls. The original 31,812 learned parameters, feature equations and normalizations are retained without optimization, calibration or checkpoint reselection. Panel size, window

Cohort	Queries	Native AP (C4)	Current native ARGMix	H038 aligned	H042 local flags
Present	182	93.78 ± 0.84	94.45 ± 1.15	95.49 ± 0.83	96.58 ± 0.60
Bronze Age	162	94.76 ± 1.18	97.71 ± 0.75	98.17 ± 0.80	99.07 ± 0.32
BAA	14	96.94 ± 1.32	98.44 ± 1.02	98.90 ± 0.89	99.02 ± 1.07
Neo	302	97.57 ± 0.27	99.30 ± 0.26	99.41 ± 0.13	99.53 ± 0.18
Yam	26	95.43 ± 1.41	99.22 ± 0.90	98.96 ± 1.31	99.86 ± 0.19
Ancient	504	96.54 ± 0.32	98.76 ± 0.31	98.98 ± 0.32	99.39 ± 0.16
All queries	686	95.81 ± 0.45	97.61 ± 0.50	98.05 ± 0.43	98.64 ± 0.26

Table 4: Cohort span accuracy (%), mean±sample SD over eight independent development simulations. Queries counts haplotypes. Ancient and all-query rows are aggregates of the preceding cohorts. BAA and Yam remain explicit despite small sizes. Bronze Age is not asserted to be identical to the published Post-Neolithic panel. Provenance: E1, E4–E5.

Condition	Prec.	Recall	F1
Native AP (C4)	3.95 ± 0.48	32.36 ± 2.06	7.03 ± 0.78
ARGMix	9.16 ± 1.49	16.41 ± 2.22	11.74 ± 1.73
H038 aligned	10.71 ± 1.21	17.99 ± 2.01	13.42 ± 1.47
H042 flags	15.21 ± 2.01	22.30 ± 1.86	18.05 ± 1.84

Table 5: Present-day boundaries at 1 kb tolerance (%), mean±sample SD across eight simulations. The positional matcher ignores transition class identity. Provenance: E1, E4–E5 and E8.

length and the pre-population-size Relate stage differ from training. Provenance: E15.

We average the two haplotypes within each person using native physical spans, pool covered chromosome spans by base pairs, and then give each person equal cohort weight. Later-minus-earlier model assignment is $+14.27$ percentage points for EHG and $+5.52$ for CHG (Figure 2). EHG is positive on every chromosome and under all ten leave-person and three leave-chromosome deletions. CHG decreases slightly on chromosomes 20 and 22, and omitting chromosome 21 changes its pooled difference to -0.68 points. Both pooled contrasts remain positive under each leave-person deletion. These outputs describe model assignments without known local ancestry truth or calibrated mixture fractions. Separate GP-dosage affinity contrasts on the shared imputation resource are negative for both proxies, using a distinct 13-person, SNP-weighted estimand. They do not independently validate the neural assignments. The frozen application supports examining model behavior on empirical genealogies, with chromosome dependence retained explicitly.

5.7 Observed Training Cost

The completed aligned H038 run records 43.6 seconds per 32-simulation pass, 1,307.9 seconds in training and 1,698.5 seconds total elapsed time. Its peak CUDA allocated memory is 21.35 GB and reserved memory is 37.16 GB in decimal units. These are measurements of that training path, excluding full data preparation and external baseline evaluation. They neither isolate a PyG speedup nor establish the cost of H042 or the other complete pipelines.

6 Discussion and Scope

The results distinguish fitted-pipeline performance from architectural invention. Local clade-change flags exceed the earlier tensor and event-MLP conditions under matched added capacity and training. H045 subsequently improves the development span score, but almost the same improvement occurs in its own-state arm. Its 0.0096-point foreign-context effect and H043's 0.0153-point neighbor-state effect both fail their practical 0.10-point gates. H050 likewise favors stable leaves in span accuracy and continuation in boundary F1, while pair-cell memory fails both registered criteria. Established tree propagation, local conditioning, pooling and genomic recurrence remain viable explanations of the observed performance.

Several limitations determine how the scores should be used. The VAL simulations have informed model selection and architecture development. A single optimizer seed cannot quantify training instability. The native comparators preserve their released methods but differ in input evidence, training units, reference choices and selection objectives. The simulated roster and the derived C4 endpoint also differ from the published ARGMix benchmark, which uses different panel, reference and evaluation conditions. Numerical comparisons with its published tables would require a separately verified recipe.

The task is conditional on one demographic simulation family and inferred observation pipeline. High cohort means do not establish robustness to genotype errors, demographic misspecification, a different reference panel or real genomes. Full-panel ARGformer and broader external comparisons remain incomplete. Separate American three-class comparisons with native ARGformer and FLARE are also unfinished, and no independent outer TEST result is reported. A literature assessment must distinguish established components from any defensible contribution before a novelty claim is formulated.

Boundary performance is a substantive remaining weakness. Improving the ability to express a transition inside a native hold addresses a geometric restriction, while improving transition recognition addresses failures where a suitable edge already exists. Neither follows automatically from higher span accuracy. Future evaluation should preserve the fixed physical-support scorer, count real short tracts, and use independent simulations for confirmation rather than tune a post-processing rule on the current boundary errors.

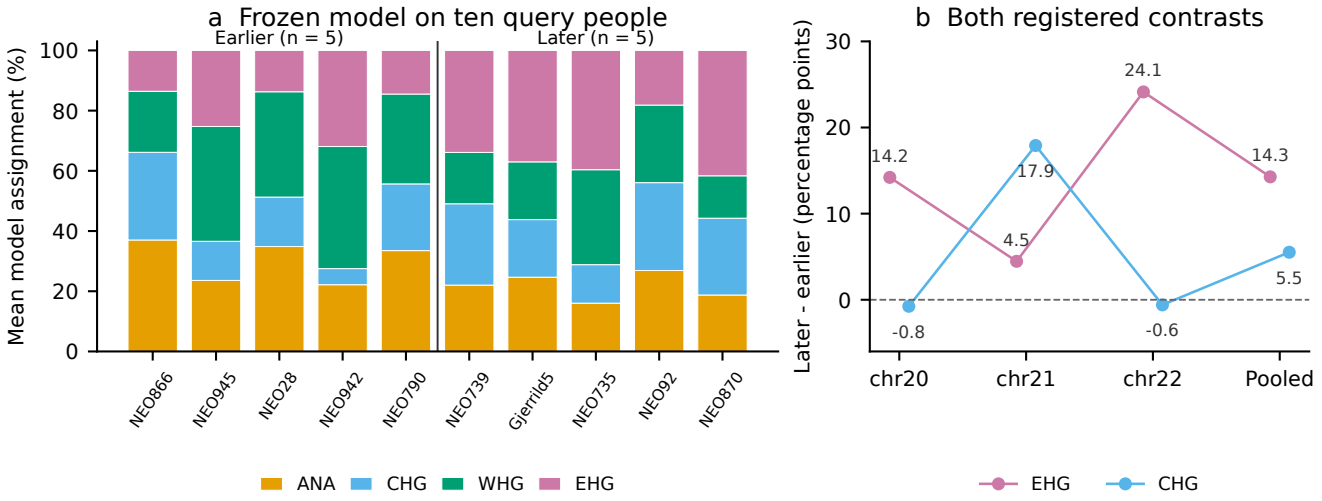


Figure 2: Frozen H045 application to the same ten Danish query people across three fixed windows. (a) Physical-span-weighted model assignments, pooled across covered chromosome spans. Each bar averages two haplotypes for one person. (b) Both registered later-minus-earlier contrasts. EHG is positive on each chromosome, while pooled CHG depends on chromosome 21. These descriptive outputs have no ancestry truth or mixture calibration.

7 Conclusion

The completed H050 continuation compares three cut-memory rules with continued training of the same whole-tree PyG and genomic-GRU model. Stable-leaf memory reaches $97.0778 \pm 0.5674\%$ present-day span accuracy, while continuation reaches $97.0000 \pm 0.4978\%$ and has higher boundary F1. Pair-cell memory fails both registered utility criteria. Together with the earlier episode controls, these development results favor testing specific structural mechanisms against continued training and simpler memory, while preserving tract-boundary metrics. They establish neither a new operator nor overall benchmark superiority. Independent assessment and matched external benchmarking remain necessary.

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A Matched Cohort Results

The same added budget and parameter count do not imply identical cohort rankings. These descriptive results supplement the registered present-day endpoint and do not redefine it.

B Completed Episode Cohort Comparison

Own-state has the larger mean in BAA, Neo, ancient and all-query aggregates, while foreign episode has the larger present-day, Bronze Age and Yam means. The small present-day difference does not establish broad cohort or metric superiority.

Cohort	Event MLP	Tensor	Local flags
Present	96.20 ± 0.70	96.27 ± 0.75	96.58 ± 0.60
Bronze Age	98.82 ± 0.44	98.83 ± 0.47	99.07 ± 0.32
BAA	99.11 ± 0.92	99.14 ± 0.89	99.02 ± 1.07
Neo	99.56 ± 0.12	99.52 ± 0.16	99.53 ± 0.18
Yam	99.76 ± 0.24	99.72 ± 0.30	99.86 ± 0.19
Ancient	99.32 ± 0.17	99.30 ± 0.21	99.39 ± 0.16
All queries	98.49 ± 0.29	98.49 ± 0.34	98.64 ± 0.26

Table 6: Matched extended-family cohort span accuracy (%), mean±sample SD over eight VAL simulations. Provenance: E2–E3 and E5.

Cohort	Own-state	Foreign episode
Present	96.82 ± 0.61	96.83 ± 0.59
Bronze Age	99.17 ± 0.40	99.20 ± 0.32
BAA	99.28 ± 1.14	99.15 ± 1.25
Neo	99.64 ± 0.09	99.61 ± 0.08
Yam	99.87 ± 0.18	99.90 ± 0.13
Ancient	99.49 ± 0.15	99.48 ± 0.13
All queries	98.78 ± 0.26	98.78 ± 0.25

Table 7: H045 cohort span accuracy (%), mean±sample SD across eight development simulations. Both arms select pass 30 after the same H042 initialization and a further 960 updates. These are descriptive cohort results, not independent validation. Provenance: E13.

C Completed Neighbor-State Control

The completed H043 comparison supplies an additional paired control, without changing the main H042 comparison. The arms compare a paired neighboring-tree state with a matched control using its own state. Each arm finishes 30 passes and 960 full-simulation updates. The true-partner arm reaches $96.5182 \pm 0.6201\%$ present-day accuracy, while the own-state control reaches $96.5029 \pm 0.6006\%$. The paired difference is 0.0153 ± 0.0920 percentage points, positive in five of eight simulations.

This effect is below the registered practical-gain criterion of 0.1 percentage points. The true-partner mean is also 0.0596 points below H042. The criterion is a practical development gate, not a significance threshold. The result does not establish a meaningful foreign-state benefit or architectural novelty. Provenance: E12. It is retained as a completed control rather than presented as a new leading method.

D Metric and Reproducibility Details

All reported accuracy cells and present-day boundary cells use eight simulations. Other diagnostic metrics can be undefined when a sparse cohort lacks a true class or boundary. Their machine-readable records retain metric-specific defined counts. Such cells are not interpreted as zero accuracy, and the tables here do not compare differently defined cohort-boundary averages.

The boundary matcher uses the fixed 1 kb tolerance and one-to-one positional matches. Its precision, recall and F1 ignore the ordered source-class transition, so a nearby wrong transition can receive positional credit. The H038 diagnostic separately records transition identity and geometric candidate competition. Its hold-edge oracle is class-agnostic and does not promise that a trained classifier can attain the geometric ceiling.

The exact-span training aggregation is appropriate because the current model is constant on each hold. It does not license the same aggregation for a model whose logits vary inside a hold. All genotype pooling and all graph/recurrent states remain observed-input computations. Target-mass arrays are neither model features nor a rule for choosing genomic windows.

The architecture description follows the exact source files listed below. The companion author README records unresolved evidence gates and compilation status. The scientific description and numerical claims are tied to completed metadata and exact source versions.

E Evidence Provenance

Tables and Figure 1 identify evidence keys below. Exact code versions, result files and full SHA256 values for E1–E15 are indexed in the companion `paper/current_full_panel_provenance.json`. E16–E17 identify the subsequent completed H050 evidence and source.

E1: H038 completed base/control scores. `experiments/results/mutation_routed_full_panel_pyg_v1_summary_root/summary.json`
`experiments/results/mutation_routed_full_panel_pyg_v1_aligned_seed101_GPU2/final_selected_full_diagnostics.json`
`experiments/results/mutation_routed_full_panel_pyg_v1_zero_concordance_seed101_GPU3/final_selected_full_diagnostics.json`

E2: H040 matched interaction comparison. `research/reviews/H040_completed_comparison_root_20261015_v1/summary.json`

E3: H042 completed local-flag comparison. `research/reviews/H042_completed_comparison_root_20261015_v1/summary.json`

E4: Native AP and ARGMix cohort diagnostics. `research/reviews/native_common_cohort_diagnostics_20261015_v1/native_cohort_metrics_mean_SD.json`

E5: H042 cohort and boundary diagnostics. `research/reviews/H042_completed_cohort_diagnostics_root_20261015_v1/summary.json`

E6: H042 exact local flags and adapter. `src/arggnn/anonymous_clade_flag_control_v1.py`

E7: Whole-tree encoding, PyG sweeps and GRU. src/arggnn/mutation_routed_whole_tree_gnn_v1.py
src/arggnn/mutation_routed_pyg_gnn_v1.py
src/arggnn/anonymous_clade_gnn_v1.py
src/arggnn/mutation_routed_full_panel_pyg_model_v1.py

E8: Fixed H038 boundary-error decomposition. research/reviews/H038_boundary_error_diagnostic_source_v1/summary.json
research/reviews/H038_boundary_error_diagnostic_source_v1/biological_decomposition.json

E9: Actual native fitting metadata. data/expanded_native_ancestralpaths_v2/run/fit/fit.json
experiments/results/expanded_native_argmix_full32TRAIN8VAL_fit_v1/fit/native_fit.json

E10: Full-panel definition and actual observation provenance. src/arggnn/prepare_expanded_full_panel_v1.py
configs/ancestralpaths_full_panel_discovery_v1.json
data/expanded_full_panel_v1/ap_expanded_train_000/generation.json
data/expanded_full_panel_v1/ap_expanded_train_000/relate/inference.json

E11: Original primary native C4 scoring. experiments/results/full_panel_H038_AP_comparison_root/summary.json
research/reviews/expanded_native_argmix_H038_score_root_v1/readonly_CPU_result_v1.json

E12: Root-confirmed completed neighbor-state control. research/reviews/H043_completed_comparison_root_20261015_v1/summary.json

E13: Completed H045 matched development evidence. research/reviews/H045_completed_comparison_root_20261015_v1/summary.json
experiments/results/H045_GPU5_episode_train_seed101_v1/completed.json
experiments/results/H045_GPU5_episode_evaluate_seed101_v1/completed.json

E14: Exact H045 episode model and frozen training configuration. src/arggnn/ordered_clade_episode_gnn_v1.py
research/reviews/H045_GPU45_episode_root_source_v1/config.json

E15: Frozen empirical application and registered contrasts. experiments/results/H056_frozen_H045_Danish_three_chromosome_CPU_v2/summary.json
experiments/results/H056_frozen_H045_Danish_three_chromosome_CPU_v2/provenance.json
research/reviews/H056_frozen_H045_Danish_application_registration_20261016_v1/protocol.root_frozen_v1.json
research/reviews/H056_three_bank_application_actual_root_20261016_v2/actual_tool_terminal.root_v1.json

E16: Completed H050 four-arm training and evaluation. experiments/results/H050_GPU0_direct_readout_train_seed101_v1/completed.json
experiments/results/H050_CPU_full192_savedfit_failedCLI_root_v1/completed.json
research/reviews/H050_CPU_evaluation_failed_CLI_successor_source_20261016_v1/root_actual_metadata_v1/actual_tool_terminal.root_v1.json

The complete saved training budget and 24 checkpoints were scored by a separate successful CPU evaluation without optimization or training replay. The original training CLI exit 1 remains recorded separately from the evaluation exit 0.

E17: H050 model and matched-memory implementation. research/reviews/H050_direct_readout_cut_memory_source_20261015_v1/direct_readout_cut_memory_v1.py
research/reviews/H050_direct_readout_cut_memory_source_20261015_v1/integration_hooks_v1.py
research/reviews/order2_cut_memory_implementation_20261015_v1/order2_cut_memory_v1.py